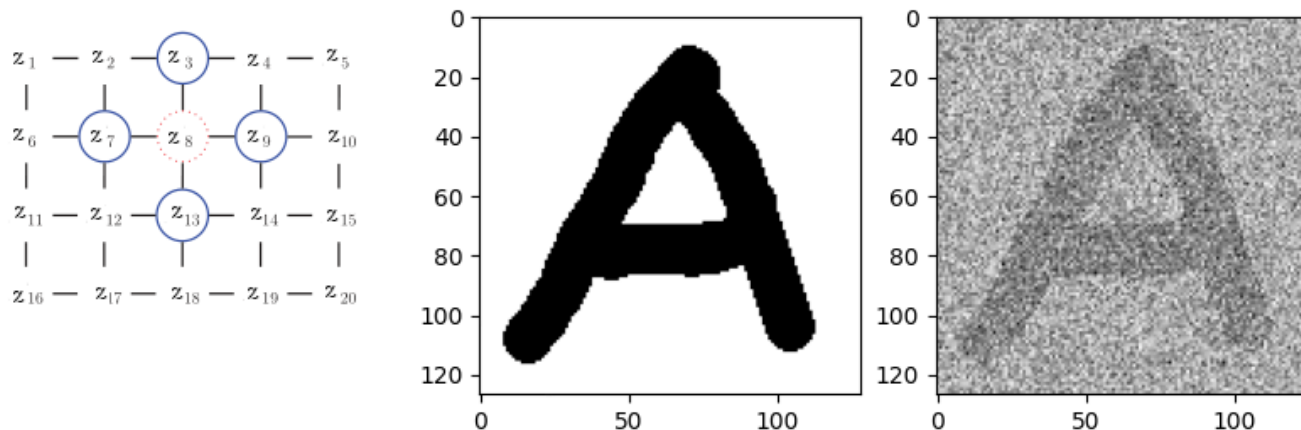


5. Sampling

- 1 Standard Distributions
- 2 Rejection Sampling
- 3 Importance Sampling
- 4 Some Practice Problems

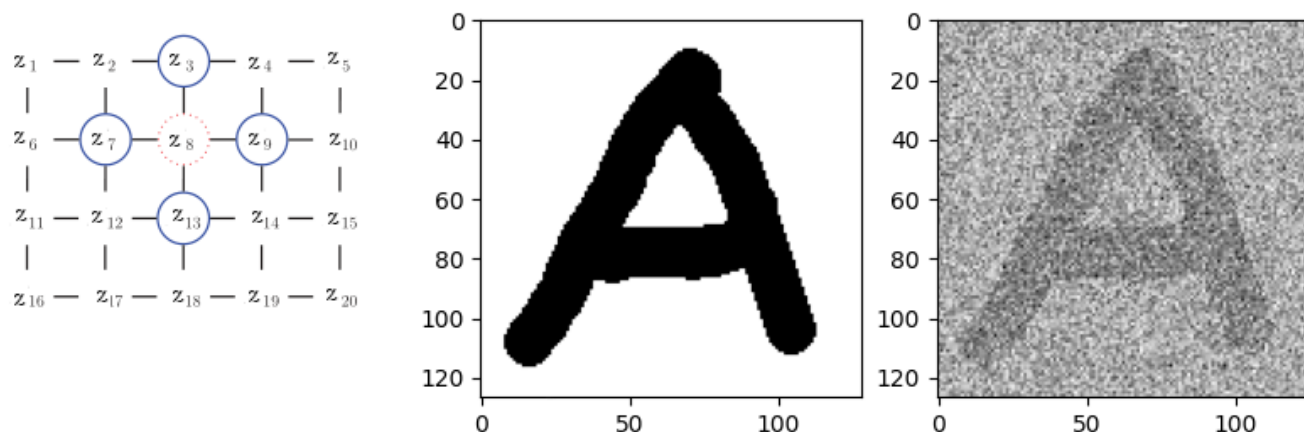
Motivation

- Image denoising: each pixel $z_j \in \{-1, 1\}$ is a hidden value of the clean image.



- Given a noisy image $\mathbf{y}$, want to recover $\mathbf{z}$. **Bayesian inference: compute the posterior $p(\mathbf{z} \mid \mathbf{y})$.**
- Likelihood: $p(\mathbf{y} \mid \mathbf{z}) = \prod_j p(y_j \mid z_j) = \prod_j \mathcal{N}(y_j \mid z_j, \sigma^2)$, i.e., each pixel is independently corrupted with Gaussian noise.

Motivation



$$p(\mathbf{z} | \mathbf{y}) = \frac{p(\mathbf{y} | \mathbf{z})p(\mathbf{z})}{p(\mathbf{y})}$$

- For good performance, need a “reasonable” prior $p(\mathbf{z})$ that models relationship between neighboring pixels.
- But this makes $p(\mathbf{z} | \mathbf{y})$ a **high dimensional** distribution, not possible to have analytical form.
- If we can sample from $p(\mathbf{z} | \mathbf{y})$, we can approximate it!

Outline

- 1 Standard Distributions
- 2 Rejection Sampling
- 3 Importance Sampling
- 4 Some Practice Problems

Introduction

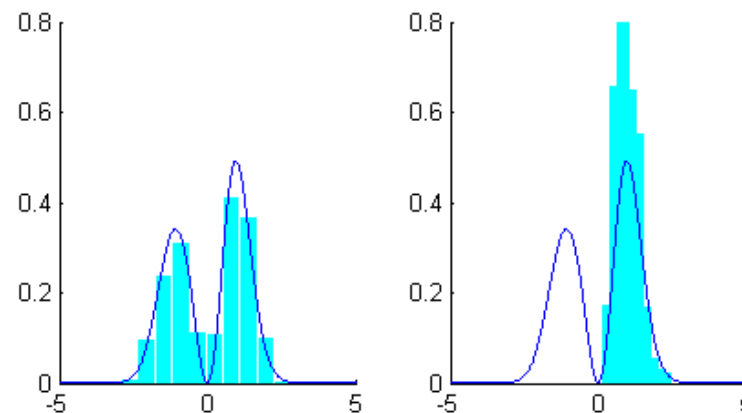
- Bayesian inference: **closed form** of the **posterior** usually only available for **conjugate priors** or exponential family likelihoods.
- Limits our modeling choices.
- How do deal with high dimensional distributions $p(\mathbf{x} \mid \boldsymbol{\theta})$?
- **Simple idea**: if we can generate **i.i.d. samples** $\mathbf{x}_1, \mathbf{x}_2, \dots \sim p(\mathbf{x} \mid \boldsymbol{\theta})$, we can use these to compute any quantity of interest, e.g., $p(\mathbf{x}_i \mid \boldsymbol{\theta})$, $p(f(\mathbf{x}) \mid \boldsymbol{\theta})$, or any

$$\mathbb{E}[f(\mathbf{x}) \mid \boldsymbol{\theta}] \approx \frac{1}{n} \sum_{i=1}^n f(\mathbf{x}_i),$$

from the SLLN.

Introduction

- We first discuss how to sample from low (1 or 2) dimensional standard distributions. Monte Carlo sampling.
- Then we discuss Markov Chain Monte Carlo techniques for high-dimensional distributions.
- The sampling methods for standard distributions are usually used as subroutines in more complex methods.
- A **sampling** procedure is **valid** for a pdf if:
 - ▶ Generate many samples from the procedure.
 - ▶ The histogram of the generated samples look like the target pdf.



Cumulative Distribution Function

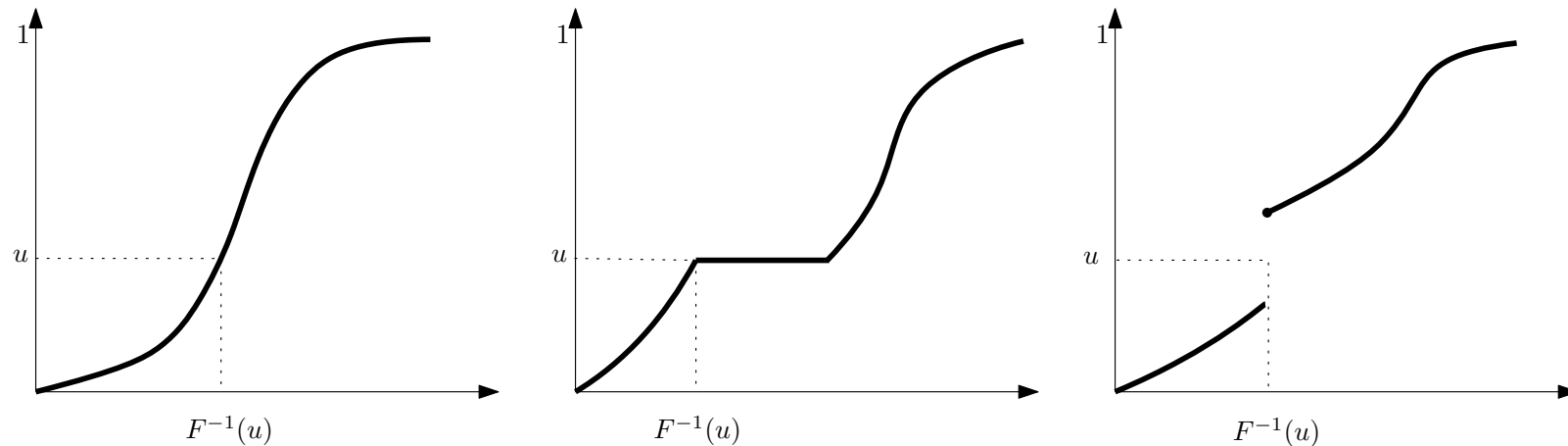
- R.v. X such that $\mathbb{P}(X \leq x) = F(x)$ is said to have cdf F .
- Let $U \sim \text{Unif}([0, 1])$ and $X = F^{-1}(U)$.

$$\begin{aligned}\mathbb{P}(X \leq x) &= \mathbb{P}(F^{-1}(U) \leq x) \\ &= \mathbb{P}(U \leq F(x)) \\ &= F(x)\end{aligned}$$

- Therefore, X is distributed according to $F(\cdot)$.
- To sample from F ,
 - ▶ sample uniform random variable U
 - ▶ apply $F^{-1}(U)$ to obtain a sample from F
- How to sample from $\text{Unif}([0, 1])$? Python: `random.random()` or `random.uniform(a, b)`

Cdf Pseudo-Inverse (Supplementary)

- R.v. X such that $\mathbb{P}(X \leq x) = F(x)$ is said to have cdf F .



- $F(x)$ may not be a bijective function. Pseudo-inverse: for $u \in [0, 1]$,

$$F^{-1}(u) = \inf\{x : u \leq F(x)\}.$$

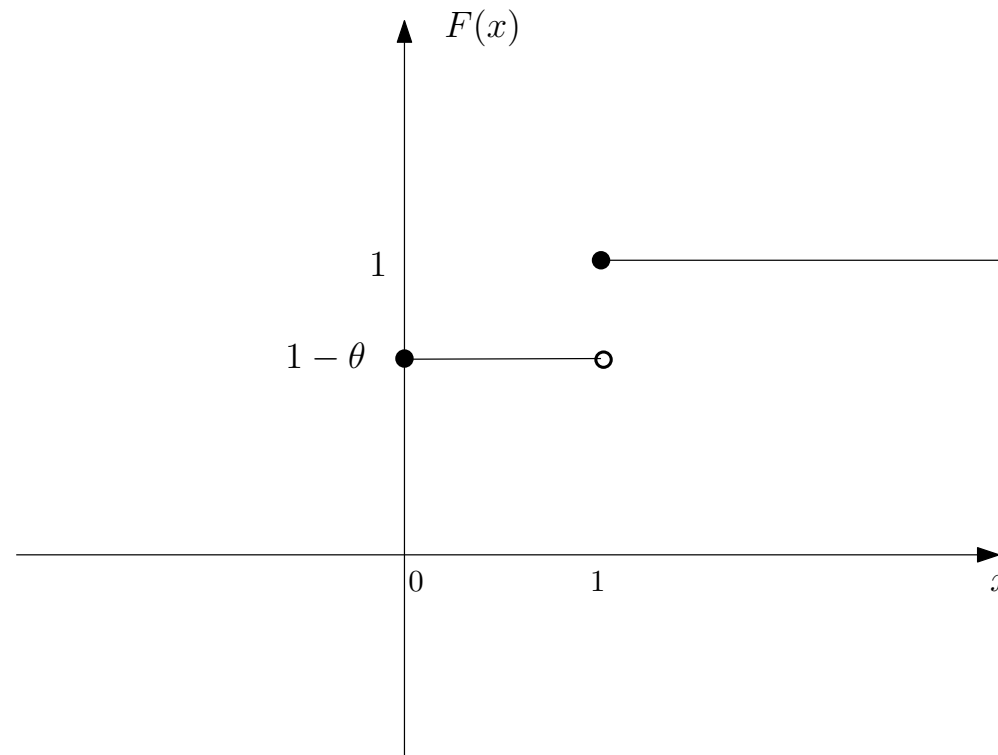
- In general, $u \leq F(x) \iff F^{-1}(u) \leq x$.

Example: Sampling Using Cdf

- Example 1: $X \sim \text{Exp}(\lambda)$, we have $F(x) = 1 - \exp(-\lambda x)$.
- In this example, we have an exact inverse F^{-1} .
- We set $X = -\frac{1}{\lambda} \log(1 - U)$, where $U \sim \text{Unif}(0, 1)$.

Example: Sampling Using Cdf

- Example 2: $X \sim \text{Bern}(\theta)$



- If $U \in [0, 1 - \theta]$, set $X = 0$. Otherwise, set $X = 1$.

Transformations

- In general, if $Y = f(X)$, then for each y ,

$$p_Y(y) = \sum_{k=1}^K \frac{p_X(x_k)}{|f'(x_k)|}, \quad (5.1)$$

where $x_1, x_2, \dots, x_K$ are solutions to the equation $f(x) = y$.

- But this requires knowing analytical form of the solutions $f(x) = y$ for any y and $f'(\cdot)$.

Transformation Examples

- $Y = aX + b$, i.e., $f(x) = ax + b$. The solution of $ax + b = y$ is $x = \frac{y-b}{a}$. We also have $f'(x) = a$.

$$\begin{aligned} p_Y(y) &= \frac{p_X(x)}{|f'(x)|} \\ &= \frac{1}{|a|} p_X\left(\frac{y-b}{a}\right). \end{aligned}$$

- If $p_X = \text{Unif}(0, 1)$, what is p_Y ?

Transformation Examples

- $Y = X^2$, i.e., $f(x) = x^2$. The solutions of $x^2 = y$ are $x_1 = \sqrt{y}$ and $x_2 = -\sqrt{y}$. We also have $f'(x) = 2x$.

$$\begin{aligned} p_Y(y) &= \frac{1}{|f'(x_1)|} p_X(x_1) + \frac{1}{|f'(x_2)|} p_X(x_2) \\ &= \frac{1}{2\sqrt{y}} (p_X(\sqrt{y}) + p_X(-\sqrt{y})). \end{aligned}$$

More Complex Distributions

- Decompose complex distributions into simple distributions, **if possible**.
- Example: Sampling from a $\text{Gamma}(a, b)$ distribution, where a is a positive integer.
 - ▶ If $Y \sim \text{Gamma}(y | a, b) = \frac{y^{a-1} e^{-by} b^a}{\Gamma(a)}$, then $Y = \sum_{i=1}^a X_i$, where $X_1, \dots, X_a$ are i.i.d. $\sim \text{Exp}(b)$.
 - ▶ Sample $X_1, \dots, X_a$ using transformation method, then sum these up to obtain sample for Y .
- What if a is not an integer? What can we do if we have more complex distributions?

Outline

- 1 Standard Distributions
- 2 Rejection Sampling
- 3 Importance Sampling
- 4 Some Practice Problems

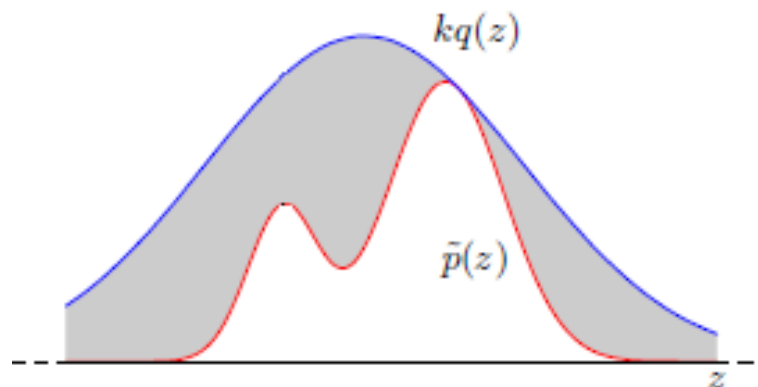
Rejection Sampling

- Sample from distributions that we know only up to a multiplicative constant,

$$p(z) = \frac{1}{M} \tilde{p}(z),$$

where M is unknown. (Recall how we compute posterior distributions.)

- Choose a proposal pdf $q(z)$ that we can sample from and a constant k such that $kq(z) \geq \tilde{p}(z)$ for all z .

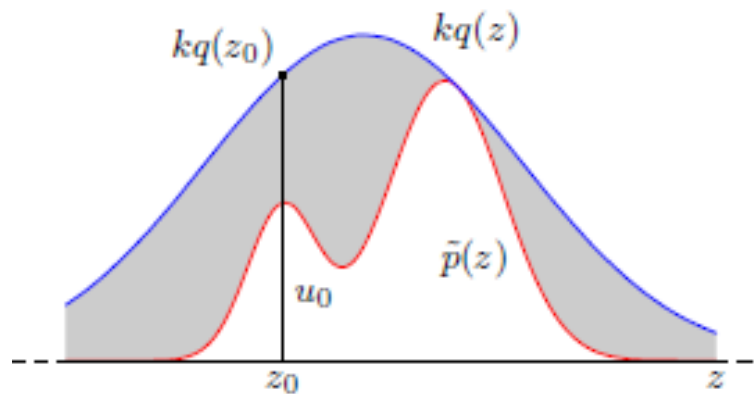


- In particular, $\text{supp } p \subset \text{supp } q$.

$$\text{supp } p = \overline{\{z : p(z) > 0\}}$$

Rejection Sampling

1. Sample z from $q(\cdot)$.
2. Sample u from $\text{Unif}[0, kq(z)]$.
3. Accept z if $u \leq \tilde{p}(z)$.



Rejection Sampling

- Acceptance probability

$$\begin{aligned}\mathbb{P}(z \text{ accepted}) &= \int \mathbb{P}(z \text{ accepted} \mid z) q(z) \, dz \\ &= \int \frac{\tilde{p}(z)}{k q(z)} q(z) \, dz \\ &= \frac{M}{k}\end{aligned}$$

- Therefore, on average takes on the order of k tries before a sample from $p(z)$ can be produced.
- Want k to be as small as possible but still need to satisfy $k q(z) \geq \tilde{p}(z)$ for all z .

Rejection Sampling

- Equivalent to sampling uniformly under the curve of $\tilde{p}(z) \Rightarrow$ accepted z is distributed according to $p(z)$.

$$\begin{aligned}\mathbb{P}(z \leq z_0 \mid z \text{ accepted}) &= \frac{\mathbb{P}(z \leq z_0, z \text{ accepted})}{\mathbb{P}(z \text{ accepted})} \\ &= \frac{\int_{z \leq z_0} \int_{u \in [0, \tilde{p}(z)]} \frac{1}{kq(z)} q(z) \, du \, dz}{M/k} \\ &= \frac{1}{M} \int_{z \leq z_0} \tilde{p}(z) \, dz,\end{aligned}$$

which is the cdf of $p(z)$.

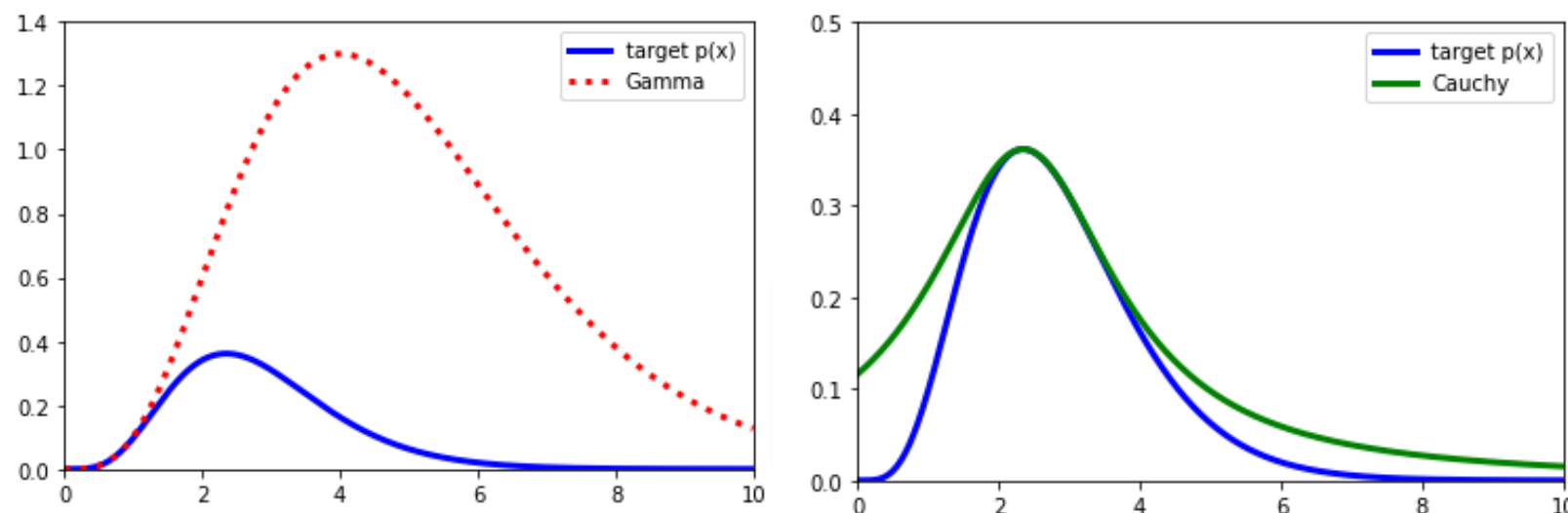
Rejection Sampling: Gamma Distribution

- Example: Sampling from $p(z) = \text{Gamma}(z \mid a, b) = \frac{b^a}{\Gamma(a)} z^{a-1} \exp(-bz)$, where $a > 1$.
- Recall if a is integer, we can sample $X_1, \dots, X_a \sim \text{Exp}(b)$ and take $Z = X_1 + \dots + X_a \sim \text{Gamma}(\cdot \mid a, b)$.
- For non-integer a , let's use $q(z) = \text{Gamma}(z \mid c, b-1)$, where $c = \lfloor a \rfloor$.

$$\begin{aligned}
 \frac{p(z)}{q(z)} &= \frac{\text{Gamma}(z \mid a, b)}{\text{Gamma}(z \mid c, b-1)} \\
 &= \frac{\Gamma(c)}{\Gamma(a)} \frac{z^{a-1} b^a \exp(-bz)}{z^{c-1} (b-1)^c \exp(-(b-1)z)} \\
 &= \frac{\Gamma(c) b^a}{\Gamma(a) (b-1)^c} z^{a-c} \exp(-z),
 \end{aligned}$$

where the maximum is attained when $z = a - c$. We choose k to be this value (as small as possible to make $p(z) \leq kq(z)$ for all z).

Rejection Sampling: Gamma Distribution



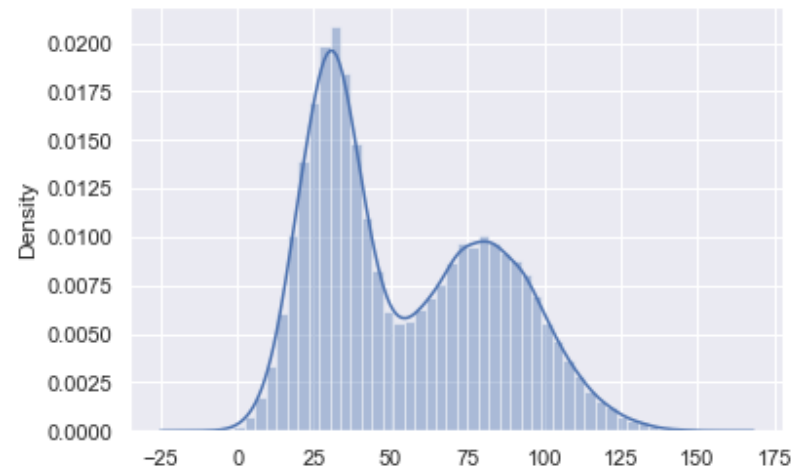
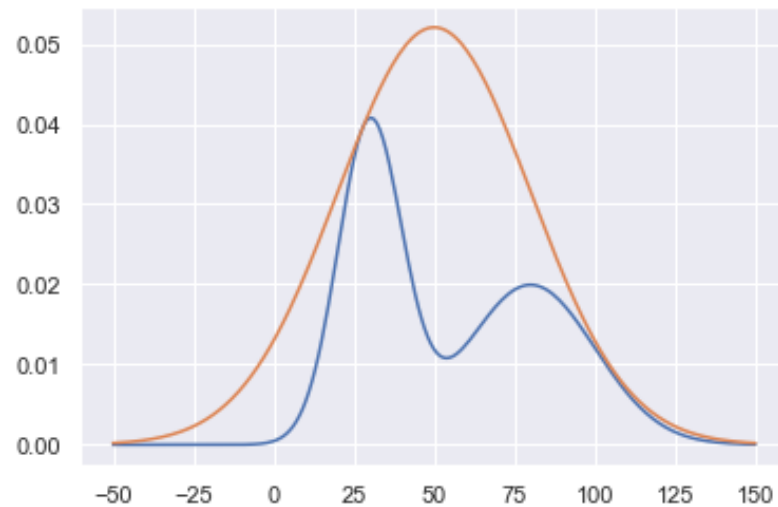
- Better choice: use **Cauchy distribution** $q(z) = \frac{1}{\pi\gamma(1+(z-c)^2/\gamma^2)}$ with $c = (a - 1)/b$ and $\gamma^2 = (2a - 1)/b^2$. It can be shown that smallest value of

$$k = \frac{\pi (a - 1)^{a-1} \sqrt{2a - 1} e^{1-a}}{\Gamma(a)}$$

- [M12]: 05_rejection_sampling_demo.ipynb

Rejection Sampling: Gaussian Mixture

05_rejection_sampling_toy.ipynb



Rejection Sampling for Bayesian Inference

- Recall in Bayesian inference, we typically require the posterior

$$p(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}.$$

- This is often not analytically tractable ($p(\mathcal{D})$ is high dimensional integral and hard to compute) unless $p(\boldsymbol{\theta})$ is a conjugate prior.
- Rejection sampling of $\tilde{p}(\boldsymbol{\theta}) = p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})$.
- If choose $q(\boldsymbol{\theta}) = p(\boldsymbol{\theta})$:

$$k = \max_{\boldsymbol{\theta}} \frac{\tilde{p}(\boldsymbol{\theta})}{q(\boldsymbol{\theta})} = \max_{\boldsymbol{\theta}} p(\mathcal{D} \mid \boldsymbol{\theta}),$$

which corresponds to the MLE of $\boldsymbol{\theta}$.

Outline

- 1 Standard Distributions
- 2 Rejection Sampling
- 3 Importance Sampling**
- 4 Some Practice Problems

Importance Sampling

- Often, we need to evaluate expectations $\mathbb{E}_p[f(z)] = \int_{\mathbb{R}} f(z)p(z) \, dz$, where $z \sim p(z)$ (e.g., in E step of EM algorithm).
- From the SLLN, we can **approximate** this as a sum, $\frac{1}{n} \sum_{i=1}^n f(z_i)$ by sampling $z_1, \dots, z_n$ from $p(z)$.
- Idea: **sample z where $|f(z)|p(z)$ is large** for better efficiency rather than from $p(z)$ directly.
- E.g., $f(z) = \mathbf{1}_{\{z \in E\}}$ where E is a *rare* event. If sample from $p(z)$ directly, may need many samples before we encounter one in E .
- We want some form of “**importance weights**”.

Importance Sampling

- Suppose $p(z)$ is difficult to sample from but $q(z)$ is easy to sample.

$$\begin{aligned}\mathbb{E}_p[f(z)] &= \int_{\mathbb{R}} f(z)p(z) \, dz = \int_{\mathbb{R}} f(z) \frac{p(z)}{q(z)} q(z) \, dz \\ &= \int_{\mathbb{R}} f(z) w(z) q(z) \, dz \\ &\approx \frac{1}{n} \sum_{i=1}^n w(z_i) f(z_i)\end{aligned}$$

where $w(z) = p(z)/q(z)$ are the **importance weights**, and $z_1, \dots, z_n \sim q(z)$ i.i.d.

Importance Sampling

- Sample $z_1, \dots, z_n$ from $q(z)$. Note that unlike rejection sampling, do **not need** $q(z) \geq p(z)$ for all z . We also do **not reject any samples**.
- Need $\text{supp } f(\cdot)p(\cdot) \subset \text{supp } q$. [Can you see why?]
- However, **convergence** performance depends on how well $q(z)$ matches $|f(z)|p(z)$, i.e., if $q(z)$ is large for z where $|f(z)|p(z)$ is large.

Example: Tail Sampling

- Want to estimate the tail probability $\mathbb{P}(X > a)$.
- **Rare event**: If sample directly from $p(x)$, may have to wait for many samples before getting one that is $> a$.
- Sample $z_1, z_2, \dots$ from a proposal distribution $q(z)$ whose support is (a, ∞) . Together with importance weights,

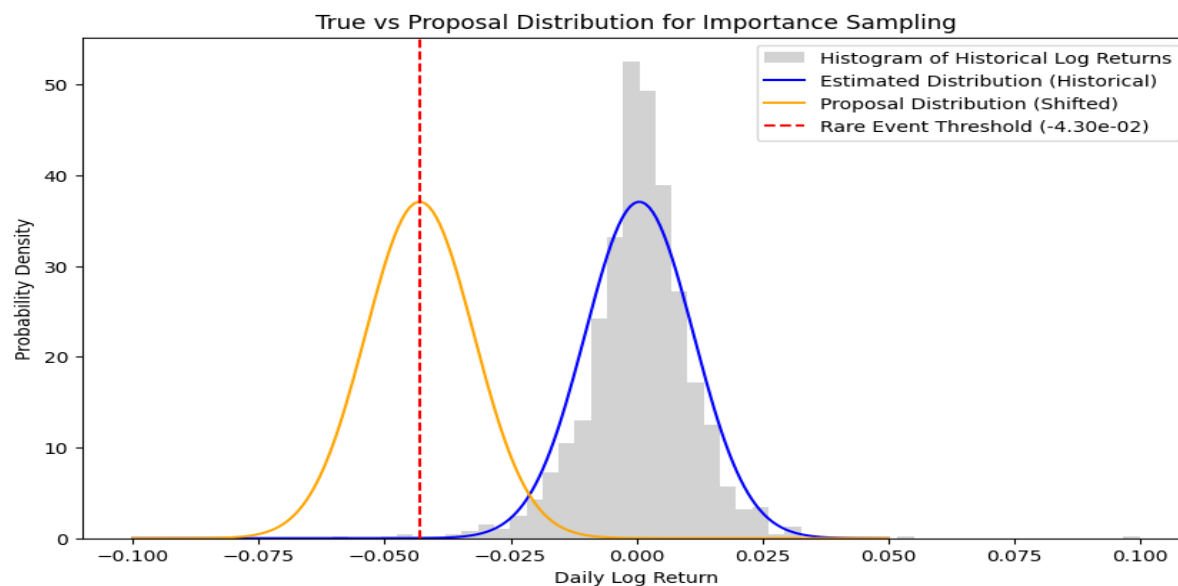
$$\mathbb{P}(X > a) \approx \frac{1}{n} \sum_{i=1}^n w(z_i)$$

- [05_importance_sampling.ipynb](#)

A Real Application

05_importance_sampling_SPY.ipynb

- Variance-reduction technique to estimate probabilities of rare events.
- Importance sampling to estimate the probability of a catastrophic drop in S&P 500 ETF (SPY).
- Let X be the log return of SPY in a day. We want to estimate $\mathbb{P}(X < a)$, where a is a large negative value, which is a rare event.



Importance Sampling

- If we only know how to compute $p(z_i)$ and $q(z_i)$ up to normalizing constants:

$$\begin{aligned}\mathbb{E}_p[f(z)] &\approx \frac{1}{n} \sum_{i=1}^n w(z_i) f(z_i) \\ &= \underbrace{\frac{1}{n} \sum_{j=1}^n w(z_j)}_{\rightarrow \mathbb{E}_q w(z) = 1} \sum_{i=1}^n w_n(z_i) f(z_i) \\ &\approx \sum_{i=1}^n w_n(z_i) f(z_i),\end{aligned}$$

where

$$w_n(z_i) = \frac{w(z_i)}{\sum_j w(z_j)} \quad (5.2)$$

are *normalized* weights. Exercise: Can you see why we only need to know unnormalized versions of p and q to compute w_n ?

Sampling Importance Resampling (SIR)

- Importance sampling: gives us weighted samples according to $f(z)$ of interest.
- What if we want unweighted samples from $p(z)$?
- Steps:
 1. Sample $z_1, \dots, z_n$ from $q(z)$.
 2. Compute weights $w_n(z_1), \dots, w_n(z_n)$.
 3. **Resample with replacement** from $\{z_1, \dots, z_n\}$ according to weights $(w_n(z_1), \dots, w_n(z_n))$.

Sampling Importance Resampling (SIR)

- Why does this give us samples from $p(z)$?
- Each sample $\tilde{z}$ is drawn from a multinomial distribution over $\{z_1, \dots, z_n\}$ with weights $(w_n(z_1), \dots, w_n(z_n))$.

$$\begin{aligned}
 \mathbb{P}(\tilde{z} \leq a \mid z_1, \dots, z_n) &= \sum_{i=1}^n w_n(z_i) \mathbf{1}_{\{z_i \leq a\}} \\
 &= \frac{\sum_{i=1}^n \mathbf{1}_{\{z_i \leq a\}} p(z_i) / q(z_i)}{\sum_{j=1}^n p(z_j) / q(z_j)} \\
 &= \frac{\frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{z_i \leq a\}} p(z_i) / q(z_i)}{\frac{1}{n} \sum_{j=1}^n p(z_j) / q(z_j)} \\
 &\xrightarrow{n \rightarrow \infty} \frac{\int_{\mathbb{R}} \mathbf{1}_{\{z \leq a\}} (p(z) / q(z)) q(z) \, dz}{\int_{\mathbb{R}} (p(z) / q(z)) q(z) \, dz} \quad (\text{almost surely}) \\
 &= \int_{z \leq a} p(z) \, dz
 \end{aligned}$$

- Note: this is true only asymptotically as $n \rightarrow \infty$.
- Typically, if we draw m samples, we take $n \gg m$.

SIR for Bayesian Inference

- Want samples from $p(\boldsymbol{\theta} \mid \mathcal{D}) = p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})/p(\mathcal{D})$.
- Take $\tilde{p}(\boldsymbol{\theta}) = p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})$. Sample $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots, \boldsymbol{\theta}_n$ from $q(\boldsymbol{\theta}) = p(\boldsymbol{\theta})$ (prior for $\boldsymbol{\theta}$).
- Compute weights

$$\begin{aligned} w_n(\boldsymbol{\theta}_i) &= \frac{\tilde{p}(\boldsymbol{\theta}_i)/q(\boldsymbol{\theta}_i)}{\sum_j \tilde{p}(\boldsymbol{\theta}_j)/q(\boldsymbol{\theta}_j)} \\ &= \frac{p(\mathcal{D} \mid \boldsymbol{\theta}_i)}{\sum_j p(\mathcal{D} \mid \boldsymbol{\theta}_j)}. \end{aligned}$$

- Resample according to weights $(w_n(\boldsymbol{\theta}_1), \dots, w_n(\boldsymbol{\theta}_n))$.

Sampling for EM

- In an example EM algorithm setup, we observe incomplete data $\mathbf{x}$ while the complete data is $(\mathbf{x}, \mathbf{z})$.
- Need to compute

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \int p(\mathbf{z} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)}) \log p(\mathbf{x}, \mathbf{z} \mid \boldsymbol{\theta}) d\mathbf{z}$$
$$\approx \frac{1}{n} \sum_{i=1}^n \log p(\mathbf{x}, \mathbf{z}_i \mid \boldsymbol{\theta})$$

where we sample $\mathbf{z}_1, \dots, \mathbf{z}_n$ from $p(\mathbf{z} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})$.

- Rejection and importance sampling are **not suitable for sampling high dimensional distributions** because of slow convergence. What can we do? MCMC methods ... after a review of Markov chains.

Summary

- Sampling of standard distributions using the uniform distribution.
- Rejection sampling
- Importance sampling
- Sampling importance resampling

Outline

- 1 Standard Distributions
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Some Practice Problems

1. For $Y = -3X$, find the relationship between p_Y and p_X .

Solution: From the pdf transformation formula, $p_Y(y) = \frac{1}{3}p_X(-y/3)$.

2. In importance sampling, suppose you can only compute $\tilde{p}(z)$ where $p(z) = \frac{1}{M}\tilde{p}(z)$ and M is unknown. Show that by replacing $w(z)$ with $\tilde{p}(z)/q(z)$, we can still perform importance sampling with the same weights $w_n(z)$ in (5.2).

Solution:

$$\frac{\frac{\tilde{p}(z_i)}{q(z_i)}}{\sum_j \frac{\tilde{p}(z_j)}{q(z_j)}} = \frac{\frac{Mp(z_i)}{q(z_i)}}{\sum_j \frac{Mp(z_j)}{q(z_j)}} = \frac{\frac{p(z_i)}{q(z_i)}}{\sum_j \frac{p(z_j)}{q(z_j)}} = w_n(z_i)$$

Some Practice Problems (cont.)

3. Let $p(x) = \text{Unif}(x \mid 0, 1)$ and $q(x) = \text{Unif}(x \mid 0, 1/2)$. Let $f(x) = x^2$. Is it true that $\mathbb{E}_q \left[\frac{f(x)p(x)}{q(x)} \right] = \mathbb{E}_p[f(x)]$?

Solution:

$$\begin{aligned} \mathbb{E}_q \left[\frac{f(x)p(x)}{q(x)} \right] &= \int_0^{1/2} \frac{f(x)p(x)}{q(x)} q(x) \, dx = \int_0^{1/2} f(x)p(x) \, dx \\ &\neq \int_0^1 f(x)p(x) \, dx = \mathbb{E}_p[f(x)]. \end{aligned}$$

Some Practice Problems (cont.)

4. Explain each line of the following code snippet, starting from Line 14. What sampling method is it implementing and what distribution is it sampling?

```
1  import numpy as np
2  import scipy.stats as st
3  sigma = 1
4
5  def p(x):
6      return np.exp(-x**2/sigma)*np.heaviside(x,1)
7
8  def q(x):
9      return st.norm.pdf(x, loc=50, scale=30)
10
11 def sampling(iter=1000):
12     samples = []
13
14     for i in range(iter):
15         z = np.random.normal(50, 30)
16         u = np.random.uniform(0, k*q(z))
17
18         if u <= p(z):
19             samples.append(z)
20
21     return np.array(samples)
```

Some Practice Problems (cont.)

Solution:

- Line 14: Loops over iter number of iterations.
- Line 15: Draws a $\mathcal{N}(50, 30)$ r.v. z .
- Line 16: Draws a $\text{Unif}(0, kq(z))$ r.v. u , where $q(z) = \mathcal{N}(z | 50, 30)$.
- Line 18: Accepts the sample z if $u \leq p(z)$.

This is rejection sampling of $p'(x) \propto p(x) = e^{-x^2} \mathbf{1}_{\{x \geq 0\}}$ using the proposal density $\mathcal{N}(x | 50, 30)$.

Some Practice Problems (cont.)

5. In finance, a “black swan event” refers to a rare event that has very low probability of happening but has major consequences if the event does happen. Therefore, it is important to test how an investment portfolio performs under such scenarios. Let $R(X)$ be the investment return, which depends on a random variable $X \sim \mathcal{N}(0, \sigma^2)$ that captures the market conditions of the investment portfolio. We want to estimate the mean return under the “black swan event” that $X > 6\sigma$. Describe a sampling strategy to achieve this.

Solution: To estimate $\mathbb{E}[R(X)\mathbf{1}\{X > 6\sigma\}]$, we can perform importance sampling using a proposal distribution $q(x)$ whose support is $(6\sigma, \infty)$ (e.g., an exponential distribution shifted by 6σ). We sample $x_1, x_2, \dots, x_n$ from this proposal distribution and compute

$$\sum_{i=1}^n w_n(x_i) R(x_i),$$

where $w_n(x_i) = w(x_i) / \sum_j w(x_j)$ and $w(x) = \mathcal{N}(x \mid 0, \sigma^2) / q(x)$.